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Abstract

State the problem you address and why it matters (1–2 sentences). Describe your approach and key contributions (2–3 sentences). Summarize your main results and their significance (1–2 sentences).

1 Introduction

Deep learning has driven remarkable progress across many domains (LeCun et al., 2015; Goodfellow et al., 2016).

While the Transformer architecture (Vaswani et al., 2017) has become ubiquitous, and residual networks (He et al., 2016) remain foundational for vision tasks, scaling these models efficiently remains an open challenge.

We propose a method that builds on pre-trained representations (Devlin et al., 2019) and is optimized using Adam (Kingma & Ba, 2014).

The main contributions of this work are:

- Contribution one.
- Contribution two.
- Contribution three.

2 Related Work

Topic A.

Topic B.

Topic C.

3 Background

Assumption 3.1 (Smoothness)**.** The loss function $\mathcal{L}(\theta)$ is L-smooth, i.e., $\|\nabla\mathcal{L}(\theta) - \nabla\mathcal{L}(\theta')\| \leq L\,\|\theta - \theta'\|$ for all θ, θ'.

Definition 3.2 (ϵ-stationary point)**.** A point θ is ϵ-stationary if $\|\nabla\mathcal{L}(\theta)\| \leq \epsilon$.

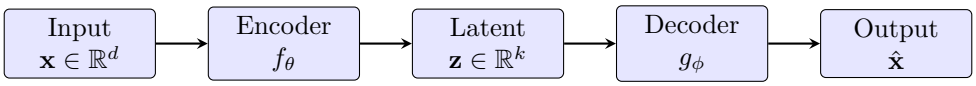

Figure 1: Model architecture overview.

Algorithm 1 Training Procedure

Require: Dataset $\mathcal{D}$, learning rate η, epochs T
Ensure: Trained parameters θ^*

```
Initialize parameters θ_0
for t = 1 to T do
    Sample mini-batch B ⊂ D
    Compute loss L(θ_t; B)
    θ_{t+1} ← θ_t − η∇_θ L(θ_t; B)
end for
return θ_T
```

4 Method

4.1 Problem Formulation

Given a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, we seek parameters θ^* that minimize:

$$\theta^* = \arg\min_\theta \frac{1}{N} \sum_{i=1}^N \mathcal{L}(f_\theta(\mathbf{x}_i), y_i) + \lambda \|\theta\|^2 \tag{1}$$

4.2 Model Architecture

4.3 Training Procedure

5 Experiments

5.1 Experimental Setup

Datasets.

Baselines.

Implementation details.

5.2 Main Results

5.3 Ablation Study

5.4 Analysis

As shown in Table 1, our method outperforms all baselines. The architecture is illustrated in Figure 1, and the training procedure is detailed in Algorithm 1. The objective in Equation (1) converges under Theorem 3.1, as proven in Theorem B.1.

[TODO: Add error bar analysis here.] **[A1: Should we include per-class breakdown?]** **[A2: Yes, moved to Section A.]**

[1] Measured on a single A100 GPU.

Table 1: Comparison with baselines on benchmark datasets. Best results are in **bold**, second-best are underlined. All metrics are averaged over 3 runs $\pm$ standard deviation.

Method	Dataset A	Dataset B	Dataset C
Baseline 1	85.2 ± 0.3	72.1 ± 0.5	91.0 ± 0.2
Baseline 2	86.7 ± 0.4	73.8 ± 0.3	91.5 ± 0.3
Baseline 3	87.3 ± 0.2	75.2 ± 0.4	92.1 ± 0.2
Ours	$\mathbf{89.1 \pm 0.2}$	$\mathbf{77.5 \pm 0.3}$	$\mathbf{93.4 \pm 0.1}$

(a) Result visualization A

(b) Result visualization B

(c) Result visualization C

Figure 2: Qualitative results. Replace placeholders with `\includegraphics[width=\linewidth]{fig.pdf}`.

We observe that performance scales log-linearly with compute budget, consistent with neural scaling laws. As shown in Figure 3, increasing model size yields diminishing returns beyond 10^{10} FLOPs. The floor of the batch size is $\lfloor N/k \rfloor$ and the ceiling is $\lceil N/k \rceil$, where N is the dataset size and k is the number of workers. We note that $\hat{\theta}_T \xrightarrow{p} \theta^*$ as $T \to \infty$ under standard regularity conditions.

Figure 3: Scaling behavior.

Key findings from our scaling analysis:

(i) Performance scales as $\mathcal{O}(\log N)$ with dataset size.

(ii) The computational cost is $\mathcal{O}\big(d^2\big)$ per layer.

(iii) Memory usage grows linearly with sequence length.

6 Discussion

Limitations.

Future work.

7 Conclusion

Broader Impact Statement

In this optional section, TMLR encourages authors to discuss possible repercussions of their work, notably any potential negative impact that a user of this research should be aware of. Authors should consult the TMLR Ethics Guidelines available on the TMLR website for guidance on how to approach this subject.

Table 2: Ablation study on Dataset A. Each row removes one component from the full model.

Configuration	**Accuracy (%)**	**Δ**
Full model	**89.1**	—
w/o Component A	87.4	−1.7
w/o Component B	88.0	−1.1
w/o Component C	86.9	−2.2

Table 3: Results broken down by model size, showing that scaling improves performance.

Model	**Size**	**Acc. (%)**	**FLOPs**
Ours	Small	87.2	1.2×10^9
	Large	**89.1**	4.8×10^{91}
Baseline	Small	85.0	1.1×10^9
	Large	86.7	4.5×10^9

Reproducibility Statement

Author Contributions

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors. Only add this information once your submission is accepted and deanonymized.

Acknowledgments

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper. Only add this information once your submission is accepted and deanonymized.

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A Additional Experimental Results

B Theoretical Results

Theorem B.1 (Convergence rate)**.** *Under Theorem 3.1, SGD with step size $\eta = 1/L$ satisfies:*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\|\nabla\mathcal{L}(\theta_t)\|^2\right] \leq \frac{2L\left(\mathcal{L}(\theta_0) - \mathcal{L}^*\right)}{T} + \frac{\sigma^2}{T} \tag{2}$$

where $\mathcal{L}^* = \inf_\theta \mathcal{L}(\theta)$ and σ^2 is the variance of the stochastic gradient.

Proof. By L-smoothness of $\mathcal{L}$ we have ... □

C Implementation Details

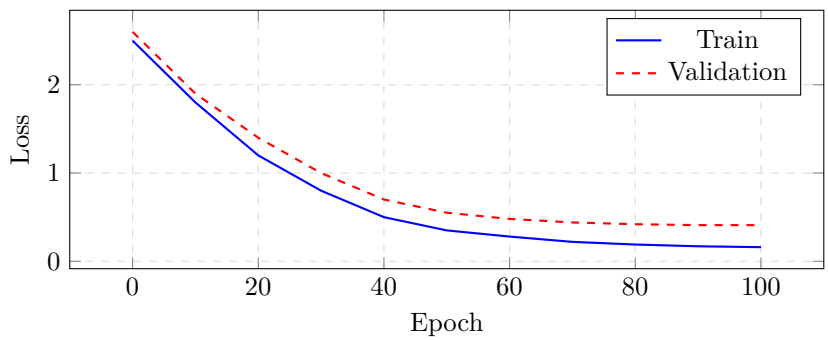

Figure 4: Training and validation loss curves.

Table 4: Hyperparameters used in all experiments.

Hyperparameter	**Value**
Learning rate	3×10^{-4}
Batch size	256
Weight decay	10^{-5}
Optimizer	AdamW
LR schedule	Cosine annealing
Warmup steps	1000
Training epochs	100
Hidden dimension	768
Number of layers	12
Attention heads	12
Dropout	0.1